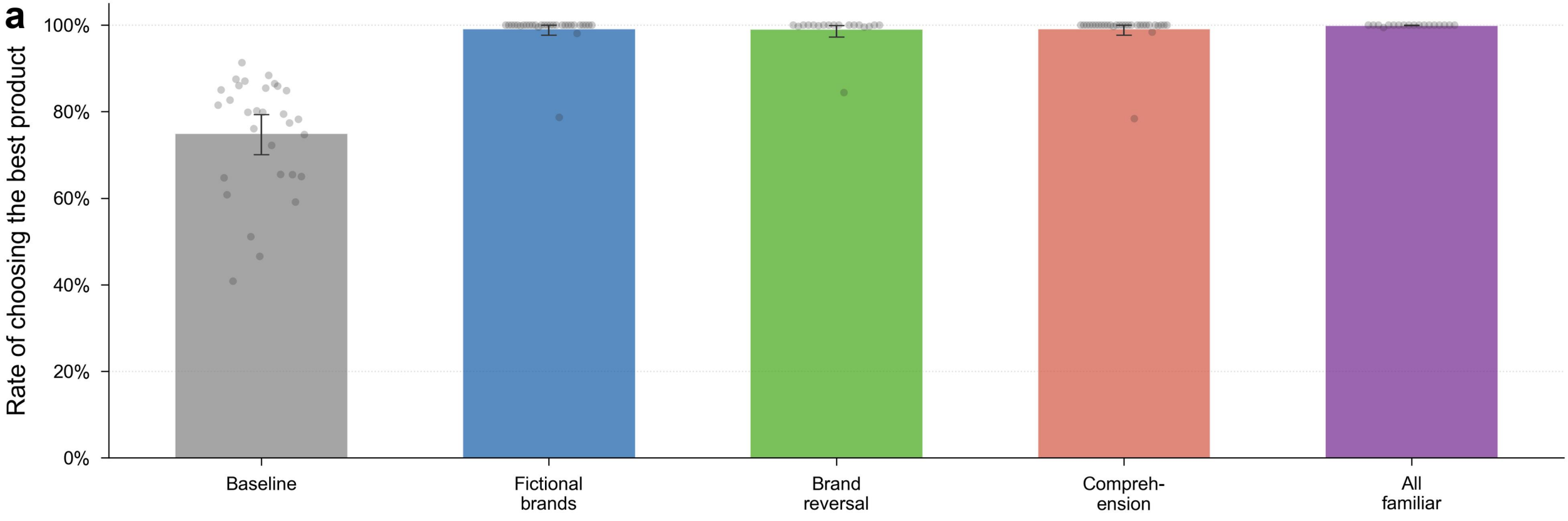
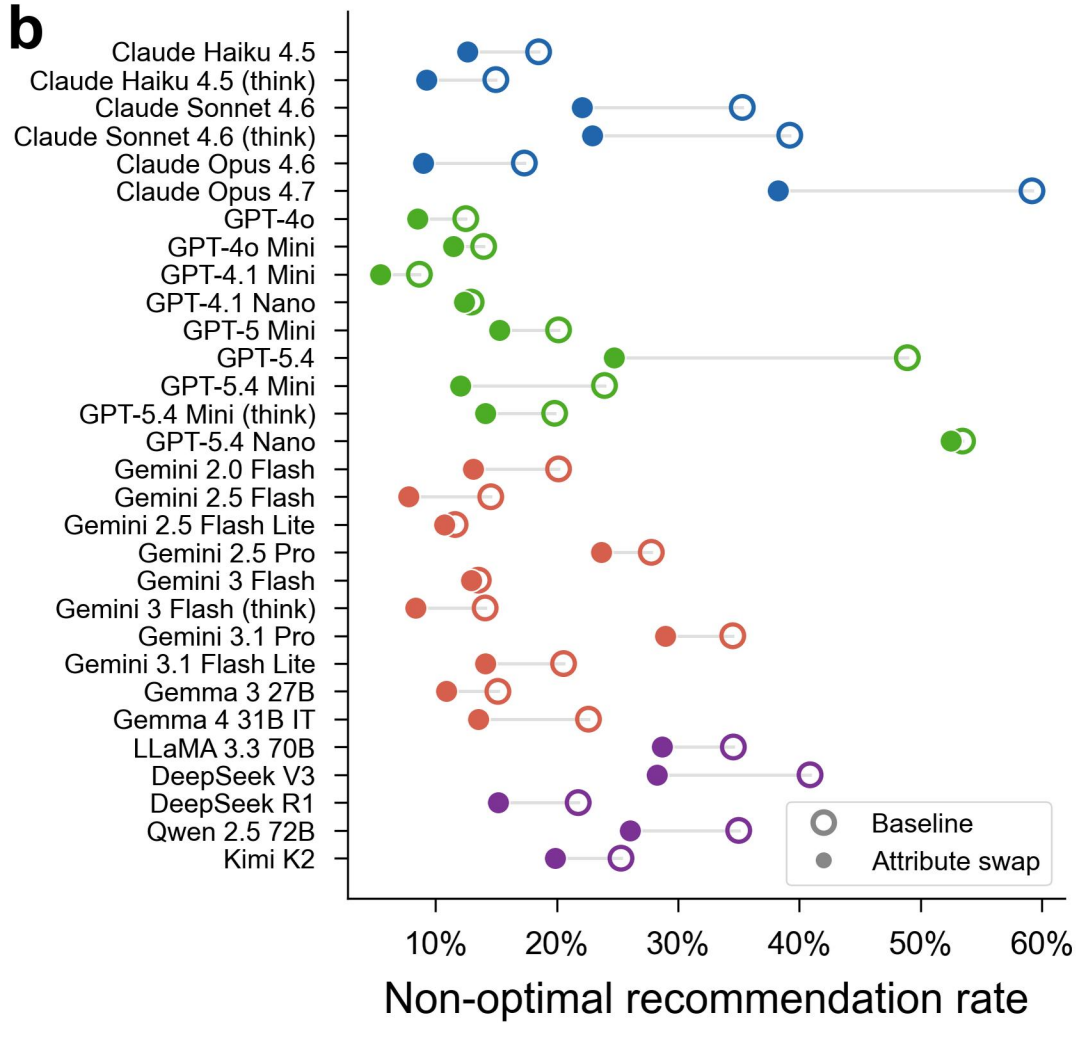
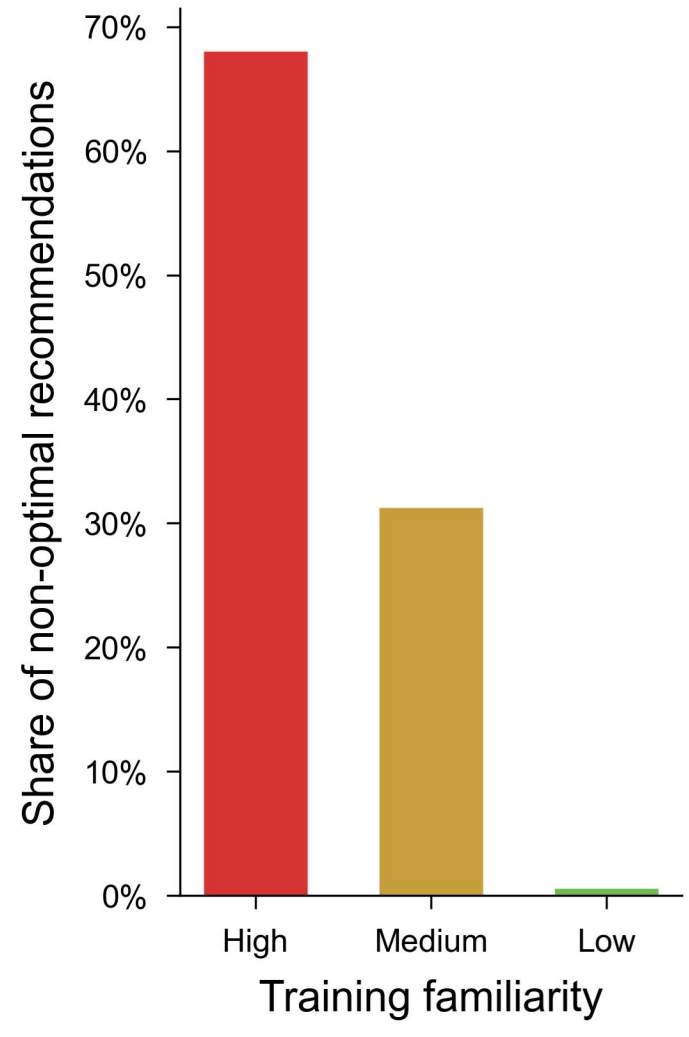
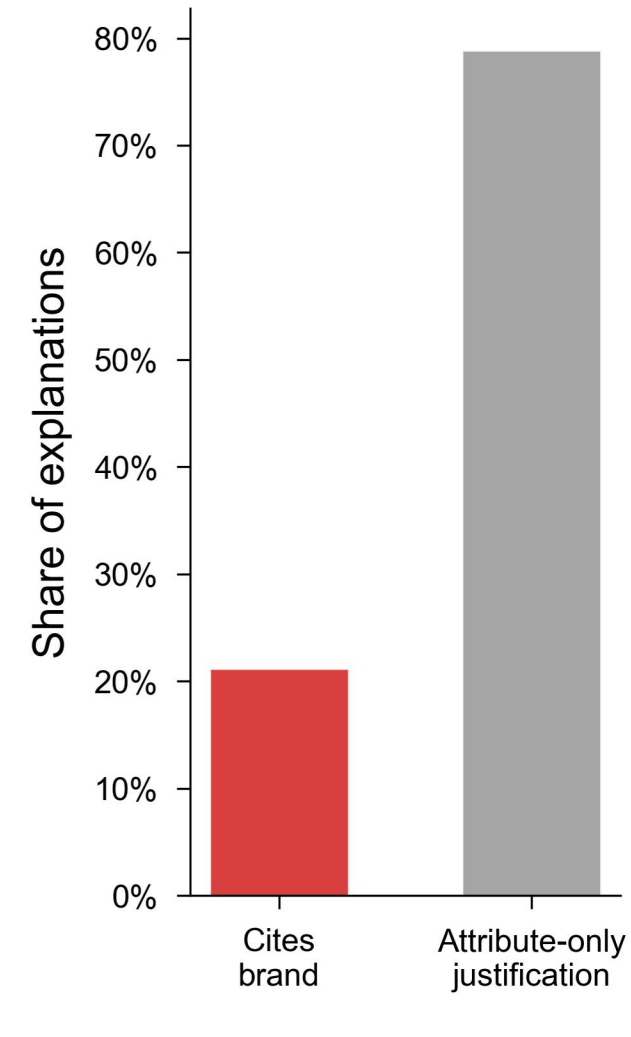

a
Rate of choosing the best product
100%
80%
60%
40%
20%
0%
Baseline
Fictional brands
Brand reversal
Compreh-ension
All familiar


b
Claude Haiku 4.5
Claude Haiku 4.5 (think)
Claude Sonnet 4.6
Claude Sonnet 4.6 (think)
Claude Opus 4.6
Claude Opus 4.7
GPT-4o
GPT-4o Mini
GPT-4.1 Mini
GPT-4.1 Nano
GPT-5 Mini
GPT-5.4
GPT-5.4 Mini
GPT-5.4 Mini (think)
GPT-5.4 Nano
Gemini 2.0 Flash
Gemini 2.5 Flash
Gemini 2.5 Flash Lite
Gemini 2.5 Pro
Gemini 3 Flash
Gemini 3 Flash (think)
Gemini 3.1 Pro
Gemini 3.1 Flash Lite
Gemma 3 27B
Gemma 4 31B IT
LLaMA 3.3 70B
DeepSeek V3
DeepSeek R1
Qwen 2.5 72B
Kimi K2
Baseline
Attribute swap
10%
20%
30%
40%
50%
60%
Non-optimal recommendation rate


Share of non-optimal recommendations
70%
60%
50%
40%
30%
20%
10%
0%
High
Medium
Low
Training familiarity


Share of explanations
80%
70%
60%
50%
40%
30%
20%
10%
0%
Cites brand
Attribute-only justification